OOD Quantification → Category-Shift Adaptation

Feature-level & sample-level weights with a cold-start gate

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Abstract

Message. Category shift is an OOD event: new labels arrive in the live stream while the reference only knows the old support. We *quantify* that OOD footprint once (RF-domain VIMP + membership), then *adapt* with two cheap weights on a CE head via `partial_fit`—no architecture change, negligible compute. A cold-start gate (N_{cap}=200 samples / new category) empowers the scarce regime and then stops, so we do not over-weight once the category is warm. On text multiclass boards, early overall-acc lift is clear (`both` +0.027 at 20 samples/cat); past the gate the curves stay slightly above uniform.

1 From OOD quantification to category-shift adapt

Problem. Reference batch REF covers known categories. Live batch LIVE mixes known support with *novel* categories (category shift / open-set arrival). The right question is not "retrain everything", but: *where is the OOD mass, and how do we give it CE budget cheaply?*

Quantify (OOD footprint). Score batch membership $W \in \{0, 1\}$ (REF vs LIVE) with a shallow RF. Feature importances $v \in \Delta^{p-1}$ are the OOD coordinates—the axes that best separate live from reference under category arrival. Restricting to the top-K of v and measuring each live point's z-distance to REF yields a membership $m_i \in [0, 1]$: high m_i = "new-category-like" in the attributed subspace. That is the quantification step; labels of novel classes, when available at adapt time, only *fuse* into m_i, they do not replace it.

Adapt (two orthogonal weights).

- **Feature-level (geometry).** $w_j^{\text{feat}} = 1 + \alpha\, \tilde{v}_j$, $x_{ij} \leftarrow x_{ij}\, w_j^{\text{feat}}$. Stretch the OOD axes so CE steps land where the shift lives.
- **Sample-level (mass).** $w_i = 1 + \lambda m_i$ (fused with a hard high-m boost), passed as `sample_weight` into `partial_fit`. Raise the CE share of new-category-like rows without dropping known-class data.

Both channels consume the *same* OOD quantification; they just act on different axes of the SGD update. Implementation is intentionally thin: one RF attribution pass, then a few `partial_fit` epochs on the live window.

Cold-start gate. Empowerment is for the scarce regime. With within-category arrival index $n_{\text{idx}}(i)$,

$$\mathcal{C} = \{i : y_i \text{ novel}, n_{\text{idx}}(i) < N_{\text{cap}}\}, \qquad N_{\text{cap}} = 200.$$

Boost only on $\mathcal{C}$; once every novel category is warm, stop feature scaling as well. Past the gate, further samples enter at uniform mass—adapt, then get out of the way.

Cost. No new backbone, no replay buffer redesign, no EWMA schedule. Compute ≈ one small RF + linear CE `partial_fit`. Suitable as an online plug-in on top of an existing head.

2 Learning curves (FULL LIVE holdout)

KPI = overall accuracy and mean novel OVA-AUC on a fixed full LIVE holdout (not a first-n subset). Budget = novel samples *per category*. Dashed line = cold-start cap at 200.

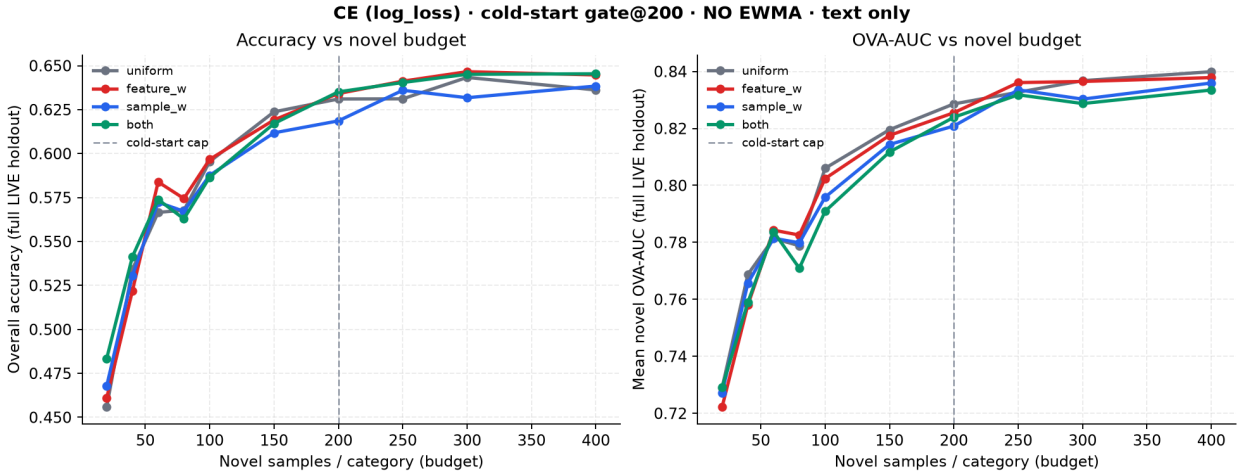


Figure 1: Accuracy (left) and novel OVA-AUC (right) vs novel samples per category. Schemes: uniform, feature-level, sample-level, both. Vertical dashed line: stop weighting after N_{cap}=200. Early regime: `both` leads (empowerment). Past the cap: weighted curves stay near / slightly above uniform—no over-boosting.

Reading Fig. 1.

- **Empowerment works.** At b=20/cat, mean overall acc rises 0.456 → **0.483** under `both` (Δ+0.027 vs uniform). Sample-only and feature-only also beat uniform; the product is strongest when the new category is scarcest.
- **Gate does its job.** After 200/cat the weighted schemes do not keep pulling away or collapsing: they track uniform and finish slightly higher (endpoint mean acc 0.636 → 0.645).
- **OVA-AUC stays flat.** Ranking quality barely moves; the gain is decision-boundary / voting quality under CE (overall acc), which is the operational KPI for cold-start deployment.

3 Endpoint table (past the gate)

Dataset	uniform	feature	sample	both
news20 (comp / rec / mix)	0.50–0.55	≈ U	≈ U	mild lift
cnae9 text	0.894	**0.916**	0.889	0.892
amazon category	0.737	**0.751**	0.747	0.744
mean overall acc	0.636	**0.645**	0.638	**0.645**
mean Δacc vs U	—	+0.009	+0.002	+0.009

Feature-level helps most when a few token dimensions carry the shift (`cnae9`, Amazon); sample-level / both help most in the early scarce window (Fig. 1, left of the dashed line).

4 Takeaway for deployment

1. **Quantify OOD, then adapt.** RF-domain VIMP + membership is the category-shift sensor; feature / sample weights are the actuator.
2. **Keep the head; use `partial_fit`.** One linear CE update with `sample_weight` (and an optional column scale) is enough—easy to ship beside an existing online pipeline.
3. **Empower, then stop.** Gate at 200 samples per new category: clear early lift, no late over-weighting, tiny compute.